# ReRAG Test Knowledge Document

A synthetic document for testing document ingestion, chunking, semantic retrieval, grounded question answering, and adaptive chunk optimization.

***Important:*** *All organizations, people, places, dates, and figures in this document are fictional and created only for software testing.*

## 1. Project Overview

The Riverbend Community Energy Pilot is a fictional community project launched by the Riverbend Planning Group in March 2024. Its purpose is to test whether shared rooftop solar installations can reduce electricity costs for public buildings and participating households. The pilot began with three sites: the North Library, the Eastside Health Centre, and the Riverbend Skills Hub. The project coordinator is Mira Sen, and the technical lead is Daniel Okafor.

The pilot has four stated goals: measure electricity generation, estimate changes in grid electricity purchases, evaluate maintenance needs, and publish a transparent quarterly report. Participation is voluntary, and household savings depend on the participant's allocation agreement and electricity use.

## 2. Equipment and Site Details

| Site | Installed capacity | Installation date | Primary use |
|---|---|---|---|
| North Library | 24 kW | April 2024 | Reading rooms and computer lab |
| Eastside Health Centre | 18 kW | May 2024 | Clinic lighting and refrigeration support |
| Riverbend Skills Hub | 30 kW | June 2024 | Training rooms and workshop equipment |

The combined installed capacity is 72 kW. The project team selected the sites because they have different daily usage patterns. The library has its highest demand in the afternoon, the health centre has a steady daytime load, and the skills hub has variable demand linked to workshop schedules. These descriptions are planning assumptions, not measurements from a real electrical grid.

## 3. Monitoring and Reporting

Each site has an energy meter that records total solar generation in kilowatt-hours (kWh). The site caretaker checks the meter display every Monday. A data officer named Leena Rao reviews the readings at the end of each month and flags missing or implausible values. The team stores monthly summaries in a shared project register.

For July through September 2024, the three sites reported a combined 8,640 kWh of solar generation. The North Library contributed 2,880 kWh, the Eastside Health Centre contributed 2,160 kWh, and the Riverbend Skills Hub contributed 3,600 kWh. These figures sum to the reported total of 8,640 kWh.

The project team estimated that grid electricity purchases across the three sites fell by 11% compared with a pre-installation baseline. This is an estimate, not a guarantee of future savings. The baseline used electricity bills from the six months before each site's installation, adjusted for the number of days in each reporting period. The report does not provide enough information to calculate a separate percentage reduction for each site.

## 4. Maintenance and Incident Log

On 12 August 2024, the North Library's monitoring gateway stopped sending data for approximately six hours. Solar generation continued during the outage according to the meter display, but the remote dashboard temporarily showed a gap. Daniel Okafor replaced a damaged network cable on 13 August. The incident was classified as a communications fault rather than a solar-panel failure.

On 4 September 2024, dust accumulation was observed on the Riverbend Skills Hub panels. The caretaker arranged cleaning on 6 September. The next monthly inspection found no visible cracks or loose mounting brackets. The log does not quantify how much generation, if any, was lost because of dust.

## 5. Roles and Responsibilities

Mira Sen coordinates schedules, community updates, and quarterly reporting. Daniel Okafor manages technical troubleshooting and installation records. Leena Rao validates monthly data and maintains audit notes. Each site caretaker checks the meter display weekly and reports visible equipment issues. The Riverbend Planning Group approves changes to the pilot's scope and publishes the quarterly report.

## 6. Budget and Next Steps

The pilot's fictional initial budget was 4.8 million credits. Of this amount, 3.1 million credits were allocated to equipment and installation, 0.9 million credits to monitoring and data systems, and 0.5 million credits to maintenance and training. The remaining 0.3 million credits were reserved for contingency expenses. These allocations add up to the full initial budget.

For the next reporting cycle, the team plans to publish a site-by-site data-quality summary, document missing meter readings, and review whether maintenance visits should occur more frequently during dusty months. The team will ask participating households for feedback about the clarity of their allocation agreements. No decision has been made to add battery storage, and no battery capacity or installation date is specified in this document.

The project team will consider the pilot successful only if its final report explains the measurement method, identifies data gaps, and separates measured results from estimates. The team has not yet published a final evaluation of long-term cost savings or emissions reductions.

## 7. Testing Notes for a Question-Answering System

**Direct retrieval examples:** Ask for the coordinator's name, the capacity of a particular site, the incident date, or the total reported generation.

**Multi-section reasoning examples:** Ask for combined installed capacity, the roles of technical troubleshooting and data validation staff, or the sum of the three sites' reported generation.

**Insufficient-information examples:** Ask for exact carbon emissions avoided, battery storage capacity, or the precise percentage of generation lost due to dust. The document does not supply these values; the system should not invent them.

*End of synthetic test document. All numbers are illustrative and should be used only to validate software behavior, not to support real-world conclusions.*